

ALLEN COUNTY AP, OH, USA

WMO: 724298

Lat: 40.708N

Lon: 84.027W

Elev: 975

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	6.7	-6.2	4.1	2.5	-1.0	5.4	7.6	28.5	25.0	26.6	28.2	8.6	240	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	90.2	73.1	87.9	72.4	85.1	71.2	76.7	85.6	75.2	83.2	73.7	81.4	10.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	130.1	81.2	72.5	124.8	79.8	71.6	120.9	78.9	40.7	85.5	39.2	83.5	37.8	81.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.4	19.0	DB	-4.4	93.6	6.1	3.9	-8.8	96.4	-12.3	98.7	-15.7	100.9	-20.2	103.7	
			WB	-5.1	79.2	5.9	2.1	-9.3	80.7	-12.8	81.9	-16.1	83.1	-20.4	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	26.8	29.4	38.9	51.3	62.0	70.5	73.5	71.7	65.7	54.1	42.6	31.9
	DBStd	18.92	12.14	11.45	11.14	9.38	8.66	6.65	5.52	5.11	7.44	9.13	9.70	10.69
	HDD50	2677	721	584	379	94	9	0	0	0	1	60	264	565
	HDD65	5748	1183	997	811	420	164	26	3	8	80	357	674	1025
	CDD50	3281	3	7	35	134	381	616	727	673	472	187	41	5
	CDD65	875	0	0	2	10	71	192	265	215	101	19	0	0
	CDH74	7321	0	0	13	90	606	1598	2336	1654	887	135	2	0
	CDH80	2252	0	0	2	9	153	524	801	471	269	23	0	0
Wind	WSAvg	9.1	11.4	11.1	10.4	10.9	9.1	8.0	6.7	6.1	6.7	8.5	10.0	10.7
Precipitation	PrecAvg	38.2	2.6	2.0	2.7	3.7	3.9	4.2	4.1	3.5	3.3	2.6	2.7	2.8
	PrecMax	53.6	7.2	5.7	5.8	9.1	6.7	12.9	10.5	9.1	6.6	5.3	5.4	7.5
	PrecMin	27.8	0.7	0.6	0.9	1.3	0.7	1.3	1.5	0.9	1.2	0.8	0.3	0.9
	PrecStd	6.7	1.3	1.2	1.1	1.5	1.8	2.3	2.1	1.7	1.6	1.3	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	65.4	74.4	81.4	88.3	92.3	94.1	91.0	90.8	82.8	71.7	61.5
		MCWB	55.4	56.3	61.8	64.7	69.4	71.8	74.6	74.2	71.1	68.4	59.4	56.0
	2%	DB	54.4	57.3	67.1	76.5	84.1	89.1	90.2	88.1	86.8	77.5	65.1	56.5
		MCWB	51.6	53.2	57.7	61.4	67.6	71.9	74.5	73.6	71.2	64.0	57.3	52.7
	5%	DB	49.3	51.7	62.2	72.5	81.0	85.9	87.7	84.8	82.5	73.1	61.9	51.8
		MCWB	46.1	47.4	54.2	59.2	66.9	70.7	73.1	72.5	68.8	62.6	54.8	48.5
	10%	DB	43.3	45.5	56.7	67.9	77.0	82.4	84.3	82.1	79.3	69.5	57.4	46.5
		MCWB	39.5	41.4	50.0	56.6	65.3	69.2	71.8	71.2	66.5	61.2	51.5	43.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	59.0	62.7	67.9	72.8	76.6	78.9	78.2	75.4	71.3	63.0	57.8
		MCDB	58.6	63.7	73.5	77.4	82.6	86.3	88.2	86.4	84.8	81.0	67.0	60.4
	2%	WB	52.1	53.3	58.9	64.0	70.4	74.8	77.2	76.3	73.2	67.8	59.1	52.6
		MCDB	54.3	56.5	65.8	72.8	80.3	84.0	86.6	84.1	82.2	74.3	63.9	55.5
	5%	WB	46.2	47.3	55.5	61.0	68.6	73.2	75.5	74.7	71.2	65.0	55.8	48.3
		MCDB	48.7	50.6	61.4	69.2	77.5	82.1	83.5	81.8	79.2	70.7	60.5	51.4
	10%	WB	39.6	41.4	50.8	58.6	66.8	71.5	74.0	73.0	69.2	61.8	52.3	43.7
		MCDB	42.7	45.4	56.1	66.7	74.9	79.8	81.3	79.5	76.1	68.5	57.4	46.5
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.2	16.6	20.1	19.6	18.7	18.9	19.0	20.8	18.5	15.6	12.3
		MCDBR	17.5	20.5	22.9	25.1	22.9	21.9	20.6	21.2	24.1	22.8	20.4	17.1
		MCWBR	15.4	17.1	15.3	14.3	11.5	9.5	8.7	9.8	11.1	12.6	14.4	15.6
	5% WB	MCDBR	17.1	19.6	20.6	22.0	19.4	18.0	17.6	17.5	19.2	18.8	18.6	16.7
		MCWBR	16.1	17.3	14.6	14.7	10.9	9.3	8.8	9.5	10.1	11.8	15.0	16.0
Clear-Sky Solar Irradiance	taub	0.307	0.325	0.347	0.399	0.440	0.444	0.445	0.442	0.398	0.357	0.324	0.301	
	taud	2.389	2.333	2.353	2.218	2.148	2.187	2.202	2.194	2.289	2.393	2.442	2.443	
	Ebn at Noon	269	280	286	277	267	265	264	261	265	266	261	263	
	Edh at Noon	26	32	36	43	48	46	45	44	37	30	25	23	
All-Sky Solar Radiation	RadAvg	508	804	1113	1486	1734	1921	1917	1721	1396	947	619	427	
	RadStd	48	94	99	131	147	141	109	75	115	82	81	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	N/A

Nomenclature: See separate page